Xiaozhen Ma

Email: shellyma626@163.com | Personal Website

EDUCATION

Southwest Jiaotong University and **University of Leeds** Sep 2023 - Jul 2027

BEng in **Electronic and Electrical Engineering** (GPA: 3.94 / 4.00. Joint Program, dual degree from both universities)

ADDITIONAL EDUCATION AND ACADEMIC PROGRAM

University of California, Irvine Jun 2026 - Aug 2026

UCInspire Summer Research Programme

St Catharine's College, Cambridge Feb 2025

Interdisciplinary Winter School (Computer Science, Data Science, and Artificial Intelligence Track)

PREPRINT

Ma, X., Tang, Z., Lai, H., Chen, R., & Jin, J. (2026). A Three-Dimensional Reverse-Projection Method for Sparse Point Cloud Completion and Its Application to High-Speed Train Nose Reconstruction. *arXiv Preprint*. https://doi.org/10.48550/arXiv.2610.05758

RESEARCH EXPERIENCES

PIKAN-Drive: Progressive Physics-Informed KANs from PilotNet Steering Regression to Fast-BEV 3D Detection

Project Lead, Supervised by Prof. Fadi Kurdahi, UC Irvine Jun 2026 - Aug 2026

- Led a progressive study of MLP, KAN, and PI-KAN architectures from PilotNet steering regression to Fast-BEV multi-camera 3D detection, establishing unified training, evaluation, and ablation pipelines.
- Integrated smoothness constraints, output-range limits, and sparsity regularization into KAN-based steering regression; systematically evaluated their combinations and weights to assess their effects on prediction accuracy.
- Built a PilotNet benchmark on 45,406 sequential Udacity samples; the best PI-KAN achieved MAE 3.209 / RMSE 6.163, reducing error by 11.3% / 9.7% relative to the MLP baseline, while the Fast-BEV KAN baseline improved mAP from 0.2559 to 0.2720 and NDS from 0.3950 to 0.4046.
- Designed nine grouped physics-informed auxiliary losses and conducted targeted ablations for Fast-BEV; showed that additional constraints did not consistently improve detection, motivating task-specific, metric-aligned, and architecture-aware supervision.

Intelligent CAD Generation and Point-Cloud-to-Editable-CAD Reconstruction

Research Assistant, Supervised by Prof. Dong-Ming Yan and Prof. Mingyang Zhao, National Laboratory of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences Feb 2026 - Present

- Deployed CAD-Assistant through the FreeCAD Python API and refined prompts with dimensional and geometric constraints to improve executable, valid CAD generation.
- Reproduced Point2CAD and ParSeNet and evaluated projection-based boundary supervision; comparative experiments showed no consistent improvement over baseline segmentation.
- Built a CAD-Recode-based pipeline that converts point clouds into executable CadQuery programs and editable CAD models, with geometric validation and STEP export.
- Investigated geometric verification, parameter refinement, and candidate reranking; the resulting 100-candidate inference pipeline reduced median Chamfer Distance by 13.4% versus a 10-candidate baseline while preserving solid validity.

Sparse Point Cloud 3D Railway Scene Reconstruction Technology Driven by Monocular Depth Estimation

Project Lead, Supervised by Prof. Zhao Tang, State Key Laboratory of Rail Transit Vehicle System Apr 2025 - Present

- Developed FrustumDense-SAM-XC3D, integrating multi-frame LiDAR, RGB images, and monocular depth; implemented viewing-frustum aggregation to densify sparse observations of railway infrastructure.
- Used LiDAR measurements to correct Depth Pro estimates through patch-wise fitting and moving least-squares residual correction, then back-projected corrected depth maps into pixel-level dense 3D point clouds.
- Combined pretrained SAM3 instance masks with cross-frame depth consistency checks to constrain object boundaries,

filter occluded observations, and fuse point clouds belonging to the same object.

- Designed class-aware radius filtering to suppress fusion noise and support category-specific extraction and visualization of tracks, buildings, and trackside facilities.

Sparse Point Cloud Completion and Surface Reconstruction of High-Speed Train Noses

Project Lead, Supervised by Prof. Zhao Tang, State Key Laboratory of Rail Transit Vehicle System Apr 2024 - Mar 2025

- Built a portable point cloud acquisition system integrating LiDAR, a camera, and an IMU, and deployed FAST-LIVO2 for sensor fusion, pose estimation, and mapping.
- Established a point-cloud processing pipeline using moving least squares for smoothing and geometric completion to address missing observations in glass-window regions.
- Proposed a completion method combining tri-axial projection, section-based interpolation, and reverse hole-boundary identification to recover missing geometry from surrounding observations.
- Compared greedy projection triangulation, Poisson reconstruction, and Marching Cubes-based reconstruction, examining window-region recovery, surface continuity, and overall shape preservation.
- Research output: Preprint posted on arXiv (arXiv:2610.05758).

HONORS AND AWARDS

- Second Prize (AI Innovation Track), 27th China Robot & Artificial Intelligence Competition - National Finals, 2025. (An Online Sports Course Evaluation System Based on Multi-Expert Collaborative Systems and Sentiment Analysis)
- Outstanding Student Leader, SWJTU-Leeds Joint School, Southwest Jiaotong University, 2025.
- Mingcheng Award, SWJTU-Leeds Joint School, Southwest Jiaotong University, 2024.

EXTRACURRICULAR ACTIVITIES AND VOLUNTEER

Academic Innovation Base of SWJTU-Leeds Joint School, Southwest Jiaotong University Sep 2023 - May 2026

Roles: Member (2023-2024); Head of Activities (2024-2025); President (2025-2026)

- As president, coordinated over 30 members and 4 departments, liaised with more than 80 guests, planned event schedules, venues, and themes, and spearheaded the development of the “Leeds Tech Month” flagship competition, attracting over 500 participating teams, with events reaching more than 3,000 faculty and students.

TECHNICAL SKILLS AND ADDITIONAL INFORMATION

Techical Skills: C/C++, Python, MATLAB, ROS2, Linux, VTK, PCL, OpenCV, VS2017, MbedStudio, OnlineGDB, Keil5.

Laguages: Mandarin(Native), English(Fluent).